# Parth Gupta

AI Engineer — Agent Harnesses & Coding Agents

hello@parthg.dev | parthg.dev | github.com/parthg1901 | linkedin.com/in/parthg1901

## Summary

AI engineer who builds production agent harnesses: agent loops, tool orchestration, prompt-cache engineering, MCP/ACP integration, and evals. Focused on making long-running agents reliable, cheap and measurable.

## Experience

**Almanak** Jun 2025 – Nov 2026

*AI Engineer (Contract)* *Remote*

- Core engineer and runtime owner on **Almanak Code**, a production agent harness (TypeScript/Bun) that designs, writes, fork-tests and backtests DeFi strategies from a chat prompt in **under 20 minutes**
- Engineered core harness systems: agent loop and tool orchestration, subagents with failure propagation, session persistence and resume, and a messages-and-parts conversation architecture
- **Cut LLM inference costs 70%** through prompt-cache engineering: stable prefix, a frozen tool list across modes (mode-switch cache hits **0%** → **91%**), and history replay that never rewrites sent context
- Bridged the agent to the web app over **ACP**; built **MCP servers** for platform actions
- Shipped human-in-the-loop flows (question widgets, awaiting-input state, mid-turn steering, message queuing) so long-running agents could pause for user decisions without losing state
- Built the eval system: headless and web-UI end-to-end runs that score strategies and catch regressions
- Agent completed **∼1,000 strategy builds** in internal evals and testing across **13 protocols and 9 chains**, from LPs and leveraged loops to delta-neutral hedges and perps
- Launched a RAG docs assistant (Cohere reranking) and a troubleshooting agent with live access to execution logs

## Projects

**Windlass** | *TypeScript, Bun; on npm* 2026

- Terminal coding agent that shows every step, token and cache hit, and runs the project's checks until they pass
- Sessions are an append-only event log, so they resume, fork and replay; `/undo` restores files and conversation
- Sandboxed commands with no network, approval for risky actions; ACP server for editors, MCP, hooks, subagents

**prefixlens** | *TypeScript, Bun; on npm* 2026

- Shows which request broke an agent's prompt cache, why, and what it cost; asserts prefix stability in CI

**Shiami** | *Python, TypeScript, LangGraph, Next.js* Jan 2025 – Jun 2025

- Designed a multi-agent assistant: a supervisor routing tasks to specialist agents via handoff tools, with RAG and web search
- Shipped user-built agent workflows (alerts, emails, transactions) on delegated wallets with an agent dashboard

## Open Source

**fastmcp** (TypeScript MCP framework): fixed `tools/list` failing for every tool when one schema can't be converted to JSON Schema (#519, shipped in 4.25.1); credited for a high-severity OAuth vulnerability (GHSA-8j7c-qjcg-7rrv)

## Education

**Maharaja Agrasen Institute of Technology** Delhi, India

*B.Tech in Artificial Intelligence and Data Science; CGPA 8.65/10* *Aug 2022 – Jun 2026*

- 16 hackathon wins, incl. two $5,000 prizes in 2024 (Filecoin track winner, Theta honorable mention)

## Technical Skills

**Languages**: TypeScript, Python, SQL, Solidity, Rust
**AI & Agents**: Agent harnesses, MCP, ACP, prompt caching, context engineering, evals, RAG (FAISS, Cohere), LangGraph; Anthropic, OpenAI, Gemini, OpenRouter APIs
**Frontend & Mobile**: React, Next.js, Tailwind CSS, shadcn/ui, Zustand, D3, Capacitor (Android), PWAs
**Backend & Infra**: Bun, Node.js, NestJS, Express, FastAPI, Prisma, Hasura (GraphQL), PostgreSQL, Redis, Docker, GCP (Cloud Run), AWS, GitHub Actions